2013 IEEE International Conference on Robotics and Automation (ICRA)
Karlsruhe, Germany, May 6-10, 2013

Flapping Wings via Direct-Driving by DC Motors

Muhammad Azhar, Domenico Campolo*, Gih-Keong Lau, Lindsey Hines, and Metin Sitti

Abstract— **In previous work, a proof-of-concept artificial flapper was devised by Campolo et al. to demonstrate that DC motors, in concert with compliant mechanisms, would be able to directly flap wings at relatively high frequencies and large angles without exceeding their operational limits. The prototype makes use of a pair of relatively long elastic strings as the compliant structures. In this article, we experimentally analyze the wing kinematics and efficiency of a more compact prototype, where small helical springs are implemented instead of elastic strings. Since the proof-of-concept prototype validated the quasi-sinusoidal assumption in spite of nonlinear aerodynamic damping, incorporating instantaneous wing kinematics into the analysis is not necessary, simplifying experiments and data processing. Along with wing kinematics and system efficiency, the possibility of controlling the wing pair to flap independently is evaluated as well.**

I. INTRODUCTION

When natural flyers hover, the torque required to actuate wing inertia can be greater than six times the torque exerted to work against aerodynamic drag [1][2]. Whereas, accelerating and decelerating the wing inertia does not play role in lift force generation. Animals with hovering flight capability overcome this exhausting condition by utilizing intrinsic elasticity in their wing actuation systems to create resonance. Most flying insects own elastic members in their thoraces for potential energy storage as well as kinetic energy release [3] and birds make use of the elasticity of their wing muscle tissues to provide restoring force [4].

In the engineering realizations of artificial flappers, up to now DC motors are still preferable among various actuator alternatives due to factors like power to weight ratio, ease of use, low in cost, and wide range in dimension. There have been efforts in the development of DC motor-driven flappers to implement the working principle of mechanical resonance [5][6][7][8]. Nevertheless, because implementing crank-arm transmissions is typical option to convert DC motors' spin to wings' reciprocation, the benefit of resonance cannot be fully harvested as the flapping actuation system is kinematically nonlinear. Furthermore, the stroke angle is unadjustable due to the geometry of crank-arm transmission and therefore tail is needed to maneuver.

This work was supported by the MINDEF-NTU-JPP/10/05 project funded by the Ministry of Defence, Singapore.

M. Azhar, D. Campolo, and G.K. Lau are with the School of Mechanical and Aerospace Engineering, Nanyang Technological University, Singapore 639798.

L. Hines is with the Robotics Institute, Carnegie Mellon University, Pittsburgh, PA 15213, USA.

M. Sitti is with the Department of Mechanical Engineering and Robotics Institute, Carnegie Mellon University, Pittsburgh, PA 15213, USA.

*Corresponding author: d.campolo@ntu.edu.sg

Avoiding the use of such transmissions in order to fully take the advantage of resonance, Campolo et al. proposed a flapping actuation concept where the wings are directly driven by DC motor [9]. The only nonlinearity that remains in this concept is the aerodynamic damping. The proof-of-concept prototype is able to construct about 120 degrees stroke angle when it resonates at the frequency of almost 40 Hz without surpassing the operational limits of the selected DC motor. Nonetheless, due to the implementation of relatively long elastic strings as the compliant structures, this prototype is unlikely functioned for autonomous flight and is meant only for characterization. In addition to the proof-of-concept prototype, another prototype which is more compact and has more potential for future development towards autonomous flight was introduced as well in brief. The latter prototype is more compact because it implements small helical springs instead of elastic strings as the compliant structures. The operations of the two prototypes are available in [10].

The objective of this work is to experimentally investigate the ability and efficiency of the compact prototype to flap the given wing at the aforementioned stroke angle and frequency in the most severe condition, i.e. maximum aerodynamic drag or damping which is achieved by fixing the wings at 90 degrees angle of attack. As the proof-of-concept prototype has demonstrated that its wings' flapping kinematics are very close to sinusoid although the aerodynamic damping is not linear, flapping angle data in time domain are no longer necessitated in the analysis and therefore the wing motion capturing process and data processing can be simplified. In addition to kinematics and efficiency, since the compact prototype makes use two DC motors side by side with each wing is attached to a DC motor, the feasibility of driving the wing pair independently for tailless maneuver is evaluated as well even though the scope of this work at the current stage does not cover its effect to the body's orientation.

II. QUASI-SINUSOIDAL ASSUMPTION FOR NONLINEAR AERODYNAMIC DAMPING

The angular position θ, velocity ω, and acceleration α of a wing undergoing reciprocation at sinusoidal regime with flapping angle amplitude $\pm\theta_0$ and flapping frequency f are respectively expressed in (1), (2), and (3), where Ω_0 is the amplitude of angular velocity and t is time.

$$\theta(t) = \theta_0 \sin(2\pi f t) \tag{1}$$

$$\omega(t) = 2\pi f \theta_0 \cos(2\pi f t) = \Omega_0 \cos(2\pi f t) \tag{2}$$

978-1-4673-5643-5/13/$31.00 ©2013 IEEE

Authorized licensed use limited to: Indian Institute of Technology - Jodhpur. Downloaded on March 31,2025 at 12:16:48 UTC from IEEE Xplore. Restrictions apply.

$$\alpha(t) = -2\,\pi\,f\,\Omega_0\,\sin(2\,\pi\,f\,t) \tag{3}$$

During sinusoidal reciprocation, the magnitude of aerodynamic drag force varies along the wing span since translational velocity is linearly proportional to the distance from center of rotation. This drag force distribution imparts nonlinear torque damping (aerodynamic damping) B to the fulcrum (DC motor's shaft) as given in (4) [11], where B_0 is the torque damping coefficient. For a given wing planform, the torque damping coefficient is formulated as in (5), where ρ is the density of the surrounding fluid, C_D stands for drag coefficient, d is the gap between wing root and fulcrum, R denotes the spanwise length of a wing, and $c(r)$ stands for chord length at distance r from the fulcrum. In the sinusoidal regime, the nonlinear aerodynamic damping expressed in (4) can be approximated by (6) (see [11] for the details). The magnitude of instantaneous power dissipated against aerodynamic drag, P_{drag}, is therefore quantified by (7).

$$B(\omega) = B_0\,\omega^2\,\text{sign}(\omega) \tag{4}$$

$$B_0 = \frac{1}{2}\,\rho\,C_D\int_d^{d+R} c(r)\,r^3\,dr \tag{5}$$

$$B(\omega) = \frac{8}{3\pi}\,B_0\,\Omega_0\,\omega \tag{6}$$

$$P_{drag} = B(\omega)\,\omega = \frac{8}{3\pi}\,B_0\,\Omega_0\,\omega^2 \tag{7}$$

To experimentally prove quasi-sinusoidal assumption, a high-speed camera was used to track the instantaneous flapping angle of the proof-of-concept prototype at various flapping angle amplitudes and frequencies [9]. One of the results is plotted in Fig. 1, indicating the wing flapping motion trajectory very closely resembles a sinusoid. Hence, the quasi-sinusoidal assumption is valid to be used in analyzing the power dissipated against nonlinear aerodynamic damping.

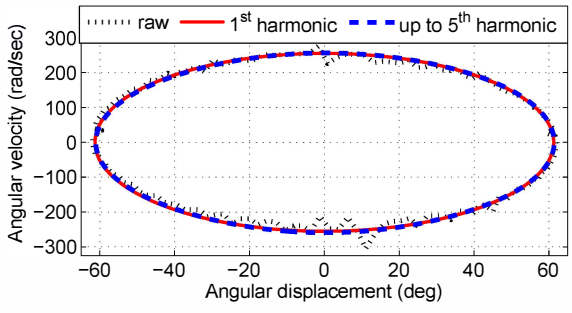


Fig. 1. State space of the wing kinematics (angular velocity vs. angular position) when the proof-of-concept prototype is supplied 2.0 V input voltage at its natural frequency of 38 Hz [9].

III. Prototype

The compact flapping wing prototype in Fig. 2 is about 6.8 grams in mass with approximately 3.2 cm height and 8 cm tip-to-tip wingspan. The components mainly consist of two brushed DC motors, two artificial wings, two pieces of helical spring, and two shaft-spring-wing couplers. The integration of each of these components constructs a pair of identical wing actuation systems which are subsequently configured side-by-side to build the complete prototype.

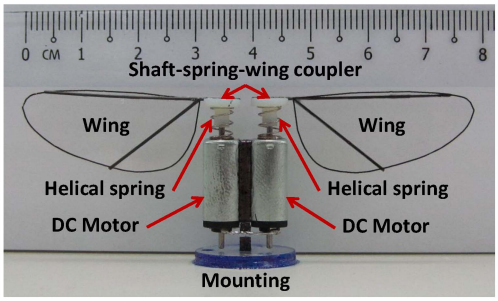


Fig. 2. Photograph of the flapping wing prototype using two DC motors driving each wing with a serial torsional spring transmission.

A. Wings

The length of each artificial wing is 30 mm, which is about 1 cm longer than the one utilized in the proof-of-concept prototype. It is composed of 102.5 micron cellulose acetate film as the membrane and two 0.5 mm carbon fiber rods as the veins. Its planform and veins' configuration are replicated from the artificial wing experimented by Zhao et al. [12]. The moment of inertia of each wing with respect to the corresponding DC motor's shaft is $J_w = 27.26\times10^{-9} kg{\cdot}m^2$.

Each wing is attached to DC motor's shaft through a shaft-spring-wing coupler as shown in Fig. 3. The wings are restricted from pitching motion and fixed at 90 degrees angle of attack to attain maximum coefficient of drag, i.e. 2.5 [12]. Applying the density of air $\rho = 1.225\,kg/m^3$, the aforementioned drag coefficient value, and geometry into (5) results in torque damping coefficient $B_0 = 4.98 \times 10^{-9}\,kg \cdot m^2 \cdot rad^{-2}$ for each wing.

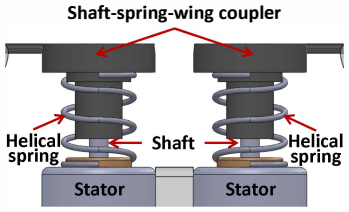


Fig. 3. CAD drawing of the motor-spring-wing connection.

B. DC Motor

We make use of small brushed-DC motor with mass of 2.5 grams (108-105 by Precision Microdrives Ltd [13]) as the source of actuation. As reported in [9], self-conducted measurements revealed that some properties of the DC motor are different from catalog as listed in Table I.

Authorized licensed use limited to: Indian Institute of Technology - Jodhpur. Downloaded on March 31,2025 at 12:16:48 UTC from IEEE Xplore. Restrictions apply.

TABLE I

SPECIFICATIONS OF PRECISION MICRODRIVES 108-105.

Source	V_{rated} (V)	I_{st} (mA)	$I_{n/l}$ (mA)	$n_{n/l}$ (rpm)	R_{mot} (Ohm)	K_t (mNm/A)
Catalog	3.0	450	46	18000	7.0	1.6
Self-measurement	3.0*	341	54	17000	8.8	1.3

Note: Magnitudes of stall current I_{st}, no-load current $I_{n/l}$, and no-load angular speed $n_{n/l}$ are correlated with rated voltage V_{rated}. R_{mot} and K_t are respectively armature resistance and torque constant of DC motor. V_{rated} in the second row is asterisked as the value was taken from catalog instead of self-measurement.

C. Elastic Structure

When the prototype flaps at an amplitude of 60 degrees, the magnitude of torque needed to actuate the artificial wing's inertia $|\tau_{J_w}(\alpha)|$ is over five times larger than the torque required to overcome the nonlinear aerodynamic damping $|B(\omega)|$. This ratio can be calculated as follows:

$$\frac{|\tau_{J_w}(\alpha)|}{|B(\omega)|} = \frac{J_w|\alpha|}{B_0\,|\omega|^2} = \frac{J_w(2\pi f)^2\theta_0}{B_0(2\pi f\theta_0)^2} = \frac{J_w}{B_0\theta_0} = 5.22.$$

The ratio above can be even higher should the inertia of the other reciprocating components such as DC motor's rotor and shaft-spring-wing coupler be taken into account. Hence, the utilization of helical spring as elastic structure to resonate with the prototype's inertia at the desired frequency is very advantageous, despite the presence of nonlinear aerodynamic damping. The lower side of the spring is mechanically grounded on the DC motor's stator and the upper side is secured to the shaft-spring-wing coupler as illustrated in Fig. 3. To ease this process, we selected spring whose both sides are flat.

Each of the wing actuation systems is mathematically expressed by second order system in (8), where J is the total inertia, b_0 is the damping coefficient due to DC motor's internal friction that can be estimated as in (9), κ is the torsion coefficient of the helical spring, and τ_m is the torque produced by the DC motor. Torsion coefficient of a system undergoing free torsional oscillation is squarely proportional to the frequency of the oscillation f as related in (10). With total inertia of each wing actuation system $J = 34.97 \times 10^{-9} kg \cdot m^2$, the torsion coefficient required to resonate with the inertia at the targetted flapping frequency of 40 Hz is $\kappa = 2.21 \cdot 10^{-3} Nm/rad$. Among the commercially available helical springs, Misumi WFH4-5 model has slightly higher linear torsion coefficient, i.e. $\kappa = 2.51 \cdot 10^{-3} Nm/rad$. This value was obtained from self-conducted static test, i.e. by longitudinally applying angular deflections to the spring and afterwards measuring the corresponding reaction torques sensed by load cell (ATI Nano17).

$$J\,\alpha + B_0\,\omega^2\,\text{sign}(\omega) + b_0\,\omega + \kappa\,\theta = \tau_m \tag{8}$$

$$b_0 = \frac{K_t\,I_{n/l}}{n_{n/l}} \tag{9}$$

$$\kappa = J\,(2\pi f)^2 \tag{10}$$

IV. EXPERIMENTS AND METHODS

A. Data Acquisition

A schematic of the experiment setup and data collection is presented in Fig. 4. Experiments were carried out by driving the two DC motors in parallel using a power amplifier and sweeping the driving frequency over a range with increment of 1 Hz, while the applied voltage was kept constant. Sensing voltage V_s and voltage applied to DC motor V_m were recorded using data acquisition (DAQ) device (National Instrument USB-6009 with 14-bit resolution) at 10 kHz sampling rate. These two voltages provide information regarding the magnitude of electric current that flows across the sensing resistor $R_s = 1.1\Omega$. This sensing resistor is actually not in series with the DC motor as shown in Fig. 4, but the current that flows through the two $22k\Omega$ resistors is negligible, i.e. $Vm/(22k\Omega + 22k\Omega) = 3V/44k\Omega = 68.2\mu A$, where 3V is the maximum operational voltage of the DC motor as specified in Table I. Therefore, the amount of electric current flowing into the DC motor I_m is very much the same as that crossing the sensing resistor.

As for wing kinematic data collection, measuring instantaneous wing flapping angle is no longer necessary since the assumption of sinusoidal wing motion was shown to be correct as elucidated in Section II. Knowing flapping angle amplitude θ_0 at each driving frequency f and input voltage V_m is adequate to carry out power analyses. Flapping angle amplitude can be easily determined from flapping motion blur image. Therefore, instead of high-speed camera, the experiment was simplified using a webcamera (PROLiNK PCC1320) to take the wing flapping motion snapshots from top as depicted in Fig. 4.

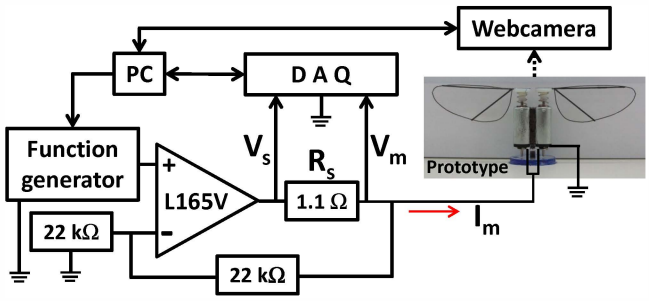


Fig. 4. Schematics of the experiment setup.

B. Data Preprocessing

The magnitude of current flowing into the DC motor I_m was obtained by subtracting V_s with V_m and then dividing the result by R_s. This raw signal of I_m contains high frequency noise and as the consequence data filtering is required. The filtering process was done by reconstructing the raw signal using Fourier series with order of harmonics $n = 5$. The time-continuous filtered signal I_m^* is expressed in (11) with t as the discrete time relative to the sampling rate of the DAQ device. Its Fourier coefficients α_n and β_n are respectively given in (12) and (13), where T quantifies the time period required to complete one full cycle.

Authorized licensed use limited to: Indian Institute of Technology - Jodhpur. Downloaded on March 31,2025 at 12:16:48 UTC from IEEE Xplore. Restrictions apply.

$$I_m^*(t) = \sum_{n=1}^{5} \alpha_n \sin(n2\pi ft) + \beta_n \cos(n2\pi ft) \tag{11}$$

$$\alpha_n = 2\,f \sum_{t=0}^{t=T} I_m(t)\,\sin(n\,2\,\pi\,f\,t) \tag{12}$$

$$\beta_n = 2\,f \sum_{t=0}^{t=T} I_m(t)\,\cos(n\,2\,\pi\,f\,t) \tag{13}$$

V. Results

A. Wing Kinematics

With input frequency swept from 33 to 53 Hz, flapping angle amplitudes are observed to peak at 42 Hz. It is a little higher than the expected value of 40 Hz and it might be caused by the utilization of slightly stiffer helical spring as mentioned in Section III. When 3 volts are applied, the acquired flapping amplitude is quite close to the target angle of 60 degrees as shown in Fig. 5 and Fig. 6.

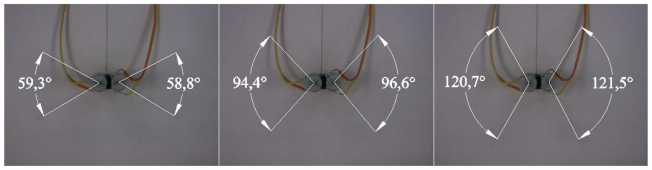


Fig. 5. Top view snapshots of stroke angles at resonant frequency of 42 Hz and input voltage of 1.0 V (left), 2.0 V (middle), and 3.0 V (right). Stroke angle is two times flapping angle amplitude.

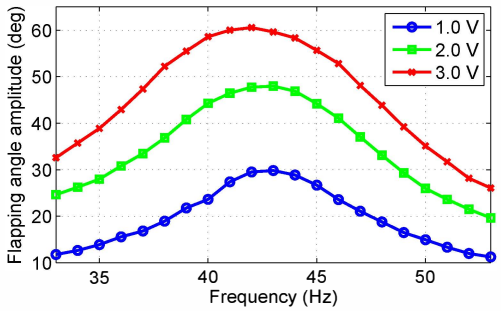


Fig. 6. Amplitudes of experimental flapping angles at various input voltages and frequencies.

B. Power and Efficiency

Based on the formula of instantaneous power dissipated against nonlinear aerodynamic damping in (7), the average power $\bar{P}_a$ over a flapping cycle is expressed as in (14). The experimental results for three input voltages are plotted in Fig. 7. Meanwhile, the average power dissipated against internal motor friction $\bar{P}_b$ is determined by (15). Unlike dissipative power, which is by definition non-negative, the instantaneous power due to inertial and elastic torques are the time derivative of the kinetic energy $\frac{1}{2}J\,\omega^2$ and the elastic energy $\frac{1}{2}\kappa\,\theta^2$, respectively. Consequently, the average power is zero. Ellington [2] defines the 'mean inertial power' as the work done by wing muscle to accelerate wings from zero to maximum angular velocity during the first half of a half-stroke as in (16). Similarly, the average power for the case of elastic torque $\bar{P}_\kappa$ is expressed as in (17).

$$\bar{P}_a = \frac{1}{2}\frac{8}{3\pi}\,B_0\,\Omega_0^3 \tag{14}$$

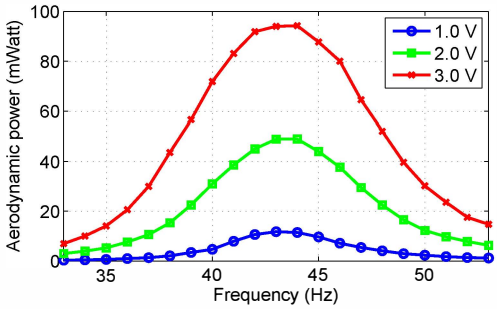


Fig. 7. Average experimental aerodynamic power dissipated by wing pair at different input voltages and frequencies.

$$\bar{P}_b = \frac{1}{2}\,b_0\,\Omega_0^2 \tag{15}$$

$$\bar{P}_J = \frac{\frac{1}{2}\,J\,\omega_{max}^2}{\frac{T}{4}} = \frac{\frac{1}{2}\,J\,\Omega_0^2}{\frac{T}{4}} \tag{16}$$

$$\bar{P}_\kappa = \frac{\frac{1}{2}\,\kappa\,\theta_{max}^2}{\frac{T}{4}} = \frac{\frac{1}{2}\,\kappa\,\theta_0^2}{\frac{T}{4}} \tag{17}$$

Dynamic efficiency is the ratio between the aerodynamic power and the summation of inertial power plus aerodynamic power [1]. Since in the case of our application we attempt to reduce the effect of inertia, the definition of dynamic efficiency needs to be modified by taking into account the contribution of elastic structure. This modification allows us to evaluate the advantage of resonance. Additionally, as the working principle of the flapping wing prototype is captured by second order system expressed in (8), the amount of power dissipated against DC motor's internal friction should be considered as well. The modified formulation of dynamic efficiency is written in (18) and the result is given in Fig. 8.

$$\eta_{dynamic} = \frac{\bar{P}_a}{\bar{P}_J + \bar{P}_a + \bar{P}_b + \bar{P}_\kappa} \tag{18}$$

The amount of input power to operate the prototype $\bar{P}_{in}$ is obtained numerically by (19) and presented in Fig. 9. To analyze how optimal the input power is consumed to overcome the nonlinear aerodynamic damping, we use overall efficiency which is defined as in (20). The result is plotted in Fig. 10.

$$\bar{P}_{in} = \frac{1}{T}\sum_{t=0}^{t=T} V_m(t)\,I_m(t)\,\Delta t \tag{19}$$

$$\eta_{overall} = \frac{\bar{P}_a}{\bar{P}_{in}} \tag{20}$$

Authorized licensed use limited to: Indian Institute of Technology - Jodhpur. Downloaded on March 31,2025 at 12:16:48 UTC from IEEE Xplore. Restrictions apply.

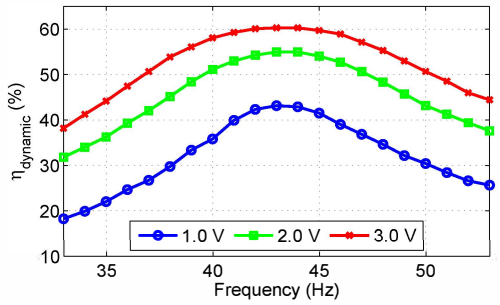


Fig. 8. Experimental dynamic efficiency at various input voltages and frequencies.

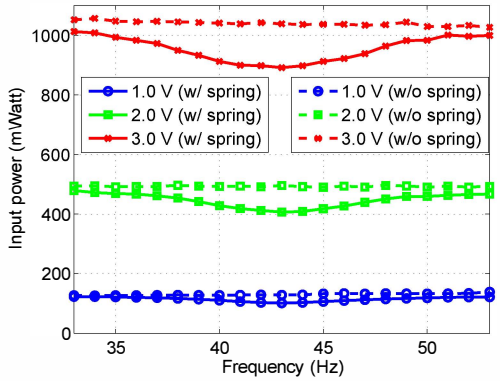


Fig. 9. Experimental input power at different input voltages and frequencies.

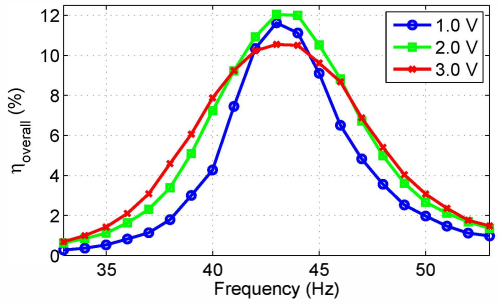


Fig. 10. Experimental overall efficiency at various input voltages and frequencies.

As the driving frequency approaches the natural frequency of the prototype, the flapping angle amplitude increases (Fig. 6) while the input power decreases (Fig. 9). Resonance is not present in the absence of helical springs. Consequently, flapping amplitude will decrease as flapping frequency is boosted and the power consumption is relatively constant over the flapping frequencies as denoted by dashed-lines in Fig. 9.

The overall efficiencies plotted in Fig. 10 are relative to maximum 50% efficiency instead of 100%. The maximum power transfer theorem for linear networks states that the maximum mechanical power delivered to the load cannot exceed the electrical losses, leading to 50% of power dissipation against the DC motor's resistance and the remaining 50% dissipated in the mechanical loads. This condition can be achieved when the equivalent mechanical impedance perfectly matches the motor electrical resistance, i.e. impedance-mismatch factor $\mu = 1$ [11].

VI. Independent Wing-beat Control and Wing Pitching Motion

The configuration of the compact prototype allows us to control the movement of the wing pair independently. In the experiment elaborated in Section IV, the two DC motors are driven parallelly using a power amplifier. In [14], the two DC motors are supplied with two separate PWM (pulse-width modulation) voltage signals generated using L293D to perform independent flapping motions and some other wings' movements. The signal can be written as in (21), where ϕ is the phase difference of the two wings and V_B is the bias voltage. Different flapping angle amplitudes of the two wings can be done by applying different V_0. Modulating V_B will shift the wing pair off the neutral position of the helical springs (Fig. 11). Unsymmetrical flapping motion of the left and right wings is achieved by adjusting ϕ.

$$V = V_0 \, sin(2\pi ft + \phi) + V_B \tag{21}$$

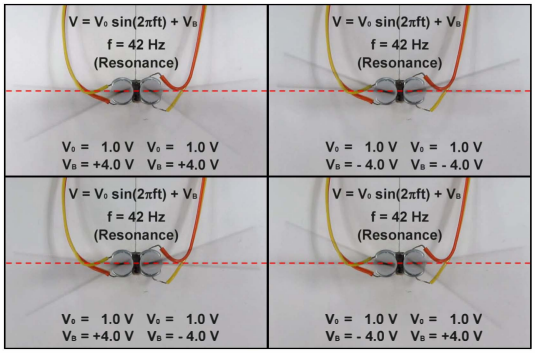


Fig. 11. Top view snapshots of off neutral axis flapping motions.

We also undertook preliminary test on wing pitching motion and observed how the artificial wing rotates using strobe light as shown in Fig. 12. We adapted passive wing rotation mechanism [15][16][17] and implemented mechanical stopper to limit the wing pitching angle at 45 degrees. The artificial wing planform used in this test is not the same as in the compact prototype and is 2 cm longer yet lighter since it is composed of 12.7 micron kapton film as the membrane and thin carbon fiber structures as the veins.

VII. Conclusion and Future Work

In comparison to the proof-of-concept prototype which is more than 20 cm in height because of the use of long elastic strings as the compliant structures, the implementation of helical springs resulted in very compact system which is more promising for autonomous flapping flight development. The compact prototype was able to drive the given wing at the desired kinematics in the most severe aerodynamic damping condition with about 60% dynamic efficiency and one-fifth of maximum overall efficiency, keeping well within

Authorized licensed use limited to: Indian Institute of Technology - Jodhpur. Downloaded on March 31,2025 at 12:16:48 UTC from IEEE Xplore. Restrictions apply.

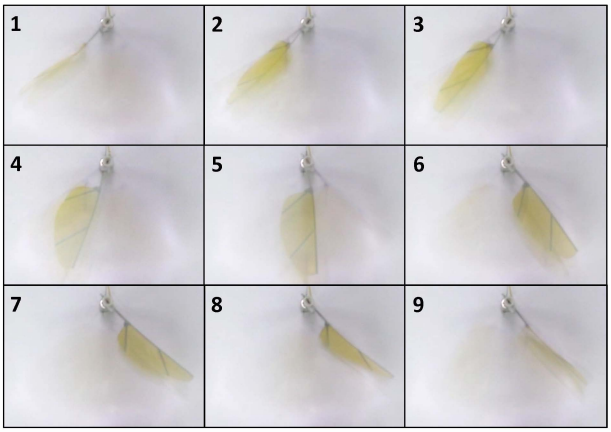


Fig. 12. Top view camera snapshots of the passive wing rotation preliminary test. The wing was driven at 23 Hz resonant frequency with stroke angle of approximately 80 degrees.

the rated limits of the selected DC motor. The same task is impossible should the helical spring is removed.

The prototype is constructed from two congruent wing actuation systems side-by-side. The combination of this physical configuration and two separate PWM signal supplies into the two DC motors enables independent control of wing beat motion as demonstrated in [14]. Unlike birds and many artificial flappers that utilize tails in their maneuvers, the independent wing motion control feature may allow the prototype to adjust its body's orientation without tail like most flying insects.

Lift and thrust productions have not been evaluated thus far as the wings were permanently set at 90 degrees angle of attack to expose the DC motors with maximum aerodynamic drag forces. Another degree of freedom allowing wing pitching motion as demonstrated in our preliminary test in Section VI is required to generate lift or thrust. This additional degree of freedom has effect on the DC motor and the second order system in (8) will be no longer sufficient. Up to which extent the motor selection method will still be valid in this new scenario shall also be scrutinized.

VIII. ACKNOWLEDGMENTS

This work was supported by the MINDEF-NTU-JPP/10/05 project funded by the Ministry of Defence, Singapore.

REFERENCES

[1] T. Weis-Fogh, "Quick Estimates of Flight Fitness in Hovering Animals, Including Novel Mechanisms for Lift Production," *J. Exp. Biol.*, vol. 59, no. 1, pp. 169–230, 1973.

[2] C.P. Ellington, "The Aerodynamics of Hovering Insect Flight. VI. Lift and Power Requirements," *Philos. T. Roy. Soc. B*, vol. 305, no. 1122, pp. 145–181, 1984.

[3] T. Weis-Fogh, "Energetics of Hovering Flight in Hummingbirds and in Drosophila, *J. Exp. Biol.*, vol. 56, no. 1, pp. 79–104, 1972.

[4] C.H. Greenewalt, "The Flight of Birds: The Significant Dimensions, Their Departure from the Requirements for Dimensional Similarity, and the Effect on Flight Aerodynamics of that Departure," *T. Am. Philos. Soc.*, vol. 65, no. 4, pp. 1–67, 1975.

[5] R. Madangopal, Z. A. Khan, and S. K. Agrawal, "Biologically inspired design of small flapping wing air vehicles using four-bar mechanisms and quasi-steady aerodynamics," *J. Mech. Design.*, vol. 127, pp. 809–816, 2005.

[6] T. Tantanawat and S. Kota, "Design of compliant mechanisms for minimizing input power in dynamic applications," *J. Mech. Des-T. ASME.*, vol. 129, pp. 1064–1075, 2007.

[7] W. Bejgerowski, A. Ananthanarayanan, D. Mueller, and S.K. Gupta, "Integrated product and process design for a flapping wing drive mechanism," it J. Mech. Design, vol. 131, pp. 0610061–0610069, 2009.

[8] S.S. Baek, K.Y. Ma, and R.S. Fearing, "Efficient Resonant Drive of Flapping-Wing Robots," in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst.*, St. Louis, MO, Oct. 11–15, 2009, pp. 2854–2860.

[9] D. Campolo, M. Azhar, G.K.Lau, and M. Sitti, "Can DC Motors Directly Drive Flapping Wings at High Frequency and Large Wing Strokes?," *IEEE/ASME Trans. Mechatronics*, in press, DOI: 10.1109/TMECH.2012.2222432, Nov. 2012.

[10] www.youtube.com/watch?v=qIkiuHg_jmg

[11] D. Campolo, "Motor Selection via Impedance-matching for Driving Nonlinearly Damped, Resonant Loads," *Mechatronics*, vol. 20, no. 5, pp. 566–573, Aug. 2010.

[12] L. Zhao, Q. Huang, X. Deng, and S.P. Sane, "Aerodynamic Effects of Flexibility in Flapping Wings," *J. R. Soc. Interface*, vol. 7, no. 4, pp. 485–497, 2009.

[13] www.precisionmicrodrives.com

[14] www.youtube.com/watch?v=ZcD2OQyVh04

[15] V. Arabagi, L. Hines, and M. Sitti, "A Simulation and Design Tool for a Passive Rotation Flapping Wing Mechanism," *IEEE/ASME Trans. Mechatronics*, vol. 18, no. 2, pp. 787–798, 2013.

[16] L. Hines, V. Arabagi, and M. Sitti, "Shape Memory Polymer based Flexure Stiffness Control in a Miniature Compliant Mechanism," *IEEE Trans. Robot.*, vol. 28, no. 4, pp. 987–990, Aug. 2012.

[17] V. Arabagi, L. Hines, and M. Sitti, "Design and Manufacturing of a Controllable Miniature Flapping Wing Robotic Platform," *Int. J. Robot. Res.*, vol. 31, no. 6, pp. 785–800, May 2012.

Authorized licensed use limited to: Indian Institute of Technology - Jodhpur. Downloaded on March 31,2025 at 12:16:48 UTC from IEEE Xplore. Restrictions apply.